

## NORMAL INCIDENCE — $\Gamma$ AND $\tau$

**Med. 1** ( $z < 0$ ): incident side. **Med. 2** ( $z \geq 0$ ): transmitted side. Boundary at  $z = 0$ .

Reflection coeff. *wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (\text{unitless})$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Intrinsic impedance *general*  $\rightarrow$  *nonmag. shortcut*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default  $\mu_r = 1$  (air, dielectric, “nonmag.”). Only use  $\mu_r \neq 1$  if problem says so (“ferromagnetic”, iron, ferrite, or  $\mu_r$  given explicitly).

Low-loss  $\eta_2$  (poor conductor)  $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon}\right) \xrightarrow{\text{drop tiny } j} \frac{\eta_0}{\sqrt{\epsilon_{r,2}}} \quad (\Omega)$$

For  $\Gamma/\tau$  just use  $\eta_0/\sqrt{\epsilon_{r,2}}$ . The  $j$  term is for accuracy with  $|\eta_c|, \theta_\eta$ .

## $\hat{E}$ PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \quad (\text{rad/m}) \quad \Rightarrow \quad k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table  $u \in \{x, y, z\} = \text{axis perp. to boundary}$

| Inc dir    | Inc           | Refl          | Trans         |
|------------|---------------|---------------|---------------|
| $+\hat{u}$ | $e^{-jk_1 u}$ | $e^{+jk_1 u}$ | $e^{-jk_2 u}$ |
| $-\hat{u}$ | $e^{+jk_1 u}$ | $e^{-jk_1 u}$ | $e^{+jk_2 u}$ |

$+\hat{u}$  direction  $\Leftrightarrow$  Med 2 at  $u \geq 0$ . Reflected sign flips, transmitted matches incident.

General template (incident has pol.  $\hat{e}$ , amplitude  $a_0$ ):

$$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp jk_1 u}, \quad \hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm jk_1 u}, \quad \hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp jk_2 u}$$

(lossy med 2: replace  $e^{\mp jk_2 u}$  with  $e^{\mp \alpha_2 u} e^{\mp j\beta_2 u}$ )

Total in medium 1:  $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$ .

$\hat{e}$  by **polarization** *plug into templates above*

| Pol.               | $\hat{e}$                               | Note              |
|--------------------|-----------------------------------------|-------------------|
| Lin. $\hat{x}$     | $\hat{x}$                               | $E_y = 0$         |
| Lin. $\hat{y}$     | $\hat{y}$                               | $E_x = 0$         |
| Lin. at $45^\circ$ | $(\hat{x} + \hat{y})/\sqrt{2}$          | in phase          |
| RHC                | $\hat{x} - j\hat{y}$                    | $\delta = -\pi/2$ |
| LHC                | $\hat{x} + j\hat{y}$                    | $\delta = +\pi/2$ |
| General            | $a_x \hat{x} + a_y e^{j\delta} \hat{y}$ | elliptical        |

$\Gamma, \tau$  act on  $\hat{e}$  unchanged. Only swap the polarization, not the formulas.

## POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted  *$\eta$ -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check:  $\%P_r + \%P_t = 100$ .

If medium 2 is denser ( $\eta_2 < \eta_1$ ),  $|\tau| > 1$  but  $\%P_t < 100\%$  because of the  $\eta_1/\eta_2$  factor.

## LOSSY MEDIA — 5 CASES

Loss tangent  $\epsilon' = \epsilon_r \epsilon_0$ ,  $\epsilon_0 \approx 1/(36\pi \times 10^9)$

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

| Type                    | $\sigma/\omega\epsilon$ | $\eta_c$ , $\alpha$ , $\beta$                                                                                                                                                                                        |
|-------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lossless ( $\sigma=0$ ) | 0                       | $\eta = \eta_0/\sqrt{\epsilon_r}$ exact; $\alpha = 0$ , $\beta = \omega\sqrt{\epsilon_r}/c$                                                                                                                          |
| Low-loss                | $< 10^{-2}$             | $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon_r}\right)$ ; $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$ , $\beta \approx \frac{\omega\sqrt{\epsilon_r}}{c}$ |
| Quasi-cond.             | $10^{-2} - 10^2$        | exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$ , see below                                                                                                                                                                 |
| Good cond.              | $> 10^2$                | $\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$ ; $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$                                                                                                               |
| Magnetic                | any                     | keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$                                                                                                                                                                           |

For  $\Gamma/\tau$ : drop  $j$  correction; use  $\eta_0/\sqrt{\epsilon_r}$ . Magnetic: keep  $\mu_r$ .

Exact (quasi-cond)  $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \quad \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon_r}$$

## SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1  $\rightarrow$  2  $\rightarrow$  3  $\rightarrow$  4*

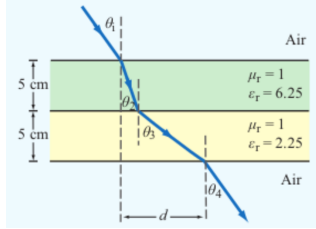
$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

*Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).*

## LATERAL DISPLACEMENT



Beam offset through  $N$  slabs *thickness  $t_i$ , refracted angle inside slab  $i$*

Slab-indexed  $\theta_i = \text{angle inside slab } i$

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

Angle-indexed (HW3 P3 style)  $\theta_1 = \text{incident}$ ,  $\theta_{i+1} = \text{inside slab } i$

$$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes  $t_1 \tan \theta_2$ , slab 2 contributes  $t_2 \tan \theta_3$ . The angle in the formula is always the one the ray makes inside that slab.

Air–slab–air: exit angle returns to  $\theta_1$  but is laterally shifted by  $d$ .

## FRESNEL OBLIQUE — $\Gamma, \tau, R, T$

Perpendicular ( $E \perp$  plane of inc.)

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad \tau_{\perp} = 1 + \Gamma_{\perp}$$

Parallel ( $E \parallel$  plane of inc.)

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad \tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$$

Critical angle (TIR) *only if  $n_1 > n_2$*

$$\sin \theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \Rightarrow \theta_i \geq \theta_c = \text{total internal refl.}$$

Brewster angle ( $\parallel$  only)  $\Gamma_{\parallel} = 0$ , *no reflection*

$$\tan \theta_B = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \quad (\text{nonmag.})$$

Reflectivity / Transmissivity *power form;  $R + T = 1$*

$$R = |\Gamma|^2, \quad T = |\tau|^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$$

Normal incidence:  $\theta_t = \theta_i = 0$ , *cos factors drop*,  $R = |\Gamma|^2$ ,  $T = |\tau|^2 (\eta_1/\eta_2)$ .

## TABLE 8-2 — $\Gamma, \tau, R, T$ SUMMARY

| Shorthand: $c_i = \cos \theta_i$ , $c_t = \cos \theta_t$ . |                                           |                                                           |                                                               |
|------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|
|                                                            | Normal ( $\theta_i = 0$ )                 | $\perp$ pol                                               | $\parallel$ pol                                               |
| $\Gamma$                                                   | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$ | $\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$     |
| $\tau$                                                     | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$             | $\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$                 |
| $\tau$ rel.                                                | $\tau = 1 + \Gamma$                       | $\tau_{\perp} = 1 + \Gamma_{\perp}$                       | $\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{c_i}{c_t}$ |
| $R$                                                        | $ \Gamma ^2$                              | $ \Gamma_{\perp} ^2$                                      | $ \Gamma_{\parallel} ^2$                                      |
| $T$                                                        | $ \tau ^2 \frac{\eta_1}{\eta_2}$          | $ \tau_{\perp} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          | $ \tau_{\parallel} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          |
| $R+T$                                                      | $T = 1 - R$                               | $T_{\perp} = 1 - R_{\perp}$                               | $T_{\parallel} = 1 - R_{\parallel}$                           |

Notes:  $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$ ;  $\eta_i = \sqrt{\mu_i / \epsilon_i}$ ; nonmag.  $\Rightarrow \eta_2/\eta_1 = n_1/n_2$ .

## FIBER OPTICS

Core index  $n_f$ , cladding  $n_c$ , surrounded by  $n_0$ . Need  $n_f > n_c$  for TIR.

Index of refraction *nonmag.*  $\mu_r = 1$

$$n = c/u_p = \sqrt{\epsilon_r}$$

Acceptance angle  $\theta_a$  *max  $\theta_i$  for TIR in fiber*

$$\sin \theta_a = \frac{1}{n_0} \sqrt{n_f^2 - n_c^2}$$

Modal dispersion (max time delay) *fiber length  $\ell$*

$$\tau = \frac{\ell n_f}{c} \cdot \frac{n_f - n_c}{n_c}$$

Max data rate  $T \geq 2\tau$  *to avoid pulse overlap*

$$f_p = \frac{1}{2\tau} = \frac{c n_c}{2\ell n_f (n_f - n_c)}$$

## RECTANGULAR WAVEGUIDE — MODES

TE mode  $E_z = 0$ ,  $H_z \neq 0$

TM mode  $H_z = 0$ ,  $E_z \neq 0$

Dimensions  $a \times b$  with  $a > b$  along  $\hat{x}, \hat{y}$ ; wave in  $\hat{z}$ .

$E_x$  form (TE) *read  $m, n$  from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on  $x = m\pi/a$ ; coef. on  $y = n\pi/b$ .

Identify  $\text{TE}_{mn}$  vs  $\text{TM}_{mn}$  *cos/sin pattern of  $E_x$  is identical for both!*

1. Read the problem statement (e.g. “a TE wave”  $\rightarrow$  TE).  
2. If  $m=0$  or  $n=0$  in the mode label  $\Rightarrow$  must be TE (TM forbids 0).

3. Source field  $H_z(x, y)$  given  $\Rightarrow$  TE. Source field  $E_z(x, y)$  given  $\Rightarrow$  TM.

TE allows at least one of  $m, n \geq 1$  (other can be 0). TM needs both  $m, n \geq 1$ . That's why  $\text{TE}_{10}$  exists but  $\text{TM}_{10}$  doesn't.

## TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand:  $C_X = \cos \frac{m\pi x}{a}$ ,  $S_X = \sin \frac{m\pi x}{a}$ ,  $C_Y = \cos \frac{n\pi y}{b}$ ,  $S_Y = \sin \frac{n\pi y}{b}$ . All fields carry an  $e^{-j\beta z}$  factor (omitted below).  $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$ .

TE mode *generator:  $\hat{H}_z$*

$$\hat{H}_z = H_0 C_X C_Y$$

$$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y$$

$$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y$$

$$\hat{E}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TE}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TE}}$$

TM mode *generator:  $\hat{E}_z$*

$$\hat{E}_z = E_0 S_X S_Y$$

$$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y$$

$$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y$$

$$\hat{H}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TM}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TM}}$$

TEM (plane wave) for reference:  $\hat{E}_x = E_0 e^{-j\beta z}$ ,  $\hat{H}_y = \hat{E}_x/\eta$ ,  $f_c$  N/A,  $k = \omega\sqrt{\mu\epsilon}$ .

## CUTOFF FREQUENCY

$f_{mn}$   $m, n$  integers  $\geq 0$  (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad (\text{Hz})$$

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} \quad (\text{m/s, guide medium})$$

Common cutoffs ( $a > b$ )

| Mode                             | $f_c$                                   |
|----------------------------------|-----------------------------------------|
| $\text{TE}_{10}$                 | $u_{p0}/(2a)$ (dominant)                |
| $\text{TE}_{01}$                 | $u_{p0}/(2b)$                           |
| $\text{TE}_{20}$                 | $u_{p0}/a = 2f_{10}$                    |
| $\text{TE}_{11}, \text{TM}_{11}$ | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$ |

Propagation requires  $f > f_{mn}$ .

## $\epsilon_r$ FROM DISPERSION

Given  $\omega$ ,  $\beta$ , mode ( $m, n$ )

$$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

Sanity:  $\beta$  in rad/m,  $a, b$  in m,  $\omega$  in rad/s.

## WAVE IMPEDANCE & $H_y$

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \quad (\Omega) \quad Z_{\text{TM}} = \eta \sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{\text{TE}}} \quad (\text{TE mode})$$

$Z_{\text{TE}} > \eta$  always (denominator  $< 1$ );  $Z_{\text{TM}} < \eta$ .

## GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \quad (\text{m/s})$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \quad (\text{m/s}) \quad \Rightarrow \quad u_p u_g = u_{p0}^2 \quad (\text{m}^2/\text{s}^2)$$

$$t = L/u_g \quad (\text{s}) \quad \text{travel time through guide of length } L$$

Higher modes have lower  $u_g \Rightarrow$  they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

$\eta_1, \eta_2$  — imp. ( $\Omega$ )  
 $\Gamma$  — refl. coeff (unitless)  
 $\tau = 1 + \Gamma$  — trans. coeff  
 $k_i = \omega \sqrt{\epsilon_{r,i}} / c$  (rad/m)  
 $\alpha_2, \beta_2$  — if med 2 lossy  
 $z = 0$  — boundary plane  
 $\eta_0 = 120\pi \approx 377 \Omega$

SNELL / OBLIQUE

$\theta_i, \theta_t$  — angles from normal  
 $n_i = \sqrt{\epsilon_{r,i}}$  — index of refr.  
 $t_i$  — slab thickness (m)  
 $d$  — lateral disp. (m)  
plane of inc. = plane of  $\hat{k}$ +normal  
 $\perp$ :  $\mathbf{E} \perp$  this plane  
 $\parallel$ :  $\mathbf{E} \parallel$  this plane

WAVEGUIDE

$a, b$  — wide, narrow dim. (m);  
 $a > b$   
 $m, n$  — mode indices (int.)  
 $f_{mn}$  — cutoff (Hz)  
 $\beta$  — prop. const (rad/m)  
 $Z_{\text{TE/TM}}$  — wave imp. ( $\Omega$ )  
 $E_x, H_y$  — field comps (V/m, A/m)  
 $\eta = \eta_0 / \sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c / \sqrt{\epsilon_r}$  — bulk phase vel.  
 $u_p = u_{p0} / \sqrt{1 - (f_c/f)^2}$   
 $u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$   
 $u_p u_g = u_{p0}^2$   
 $t = L / u_g$  — travel time  
 $c = 3 \times 10^8$  m/s

POWER & UNITS

$\%P_r = 100|\Gamma|^2$   
 $\%P_t = 100|\tau|^2 \eta_1 / \eta_2$   
 $\%P_r + \%P_t = 100$   
 $\sigma$  (S/m),  $\epsilon$  (F/m),  $\mu$  (H/m)  
 $\epsilon_0 = 8.85 \times 10^{-12}$  F/m  
 $\mu_0 = 4\pi \times 10^{-7}$  H/m  
 $1 / (36\pi) \times 10^{-9} \approx \epsilon_0$

DIPOLE TYPE BY LENGTH

| Check first <i>compute <math>\lambda = c/f</math>, then <math>l/\lambda</math></i> |                  |                     |
|------------------------------------------------------------------------------------|------------------|---------------------|
| $l/\lambda$                                                                        | Type             | Use                 |
| $< 1/50$                                                                           | Hertzian (short) | Hertzian formula    |
| $= 1/4$                                                                            | Quarter-wave     | Arbitrary formula   |
| $= 1/2$                                                                            | Half-wave        | $D = 1.64$ , simple |
| other                                                                              | Arbitrary        | Arbitrary formula   |

*If  $l/\lambda < 1/50$  stop and use Hertzian only — don't subst. small  $l$  into arbitrary formula.*

HERTZIAN DIPOLE —  $S(R, \theta)$

|                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|
| Power density <i><math>l/\lambda &lt; 1/50</math>; far field</i>                                                  |
| $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32 \pi^2 R^2} \sin^2 \theta \text{ (W/m}^2\text{)}$                   |
| $k = 2\pi/\lambda \text{ rad/m; } I_0 \text{ peak current (A); } l \text{ length (m); } R \text{ obs. dist (m).}$ |
| Max direction $\theta = \pi/2$ (broadside)                                                                        |
| Total radiated power <i>integrate over <math>4\pi \text{ sr}</math></i>                                           |
| $P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left( \frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$                        |
| Directivity <i>Hertzian, exact</i>                                                                                |
| $D_{\text{Hertz}} = 1.5 \text{ (i.e. 1.76 dB)}$                                                                   |

ARBITRARY-LENGTH DIPOLE —  $S(\theta)$

|                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power density at distance $R$ <i>any <math>l</math></i>                                                                                                   |
| $S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$         |
| Normalized pattern $F(\theta) = S(\theta)/S_{\text{max}}$ , <i>dimensionless</i>                                                                          |
| Half-wave $l = \lambda/2 \quad \pi l/\lambda = \pi/2$                                                                                                     |
| $S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}, \quad D = 1.64, \quad R_{\text{rad}} = 73 \Omega$          |
| Quarter-wave $l = \lambda/4 \quad \pi l/\lambda = \pi/4, \cos(\pi/4) = \sqrt{2}/2$                                                                        |
| $\frac{S(\theta)}{S_0} = \left[ \frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$ |
| $S_{\text{max}} = S(\pi/2) = S_0 \left( 1 - \frac{\sqrt{2}}{2} \right)^2 \approx 0.0858 S_0$                                                              |

BEAM PARAMETERS — HPBW,  $\Omega_p, D$

|                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1. Normalize                                                                                                                            |
| $F(\theta) = S(\theta)/S_{\text{max}}$                                                                                                  |
| 2. Half-power beamwidth (HPBW) <i>solve <math>F(\theta_{1/2}) = 0.5</math></i>                                                          |
| $\beta = 2\theta_{1/2} \text{ (symmetric beams)}$                                                                                       |
| example $F(\theta) = e^{-20\theta^2} \Rightarrow \theta_{1/2} = \sqrt{-\ln 0.5/20} = 0.186 \text{ rad} \Rightarrow \beta = 21.31^\circ$ |
| 3. Pattern solid angle <i>(sr)</i>                                                                                                      |
| $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$                                                         |
| If $F$ has no $\phi$ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$                                                |
| 4. Directivity                                                                                                                          |
| $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$                                                                 |
| 5. Gain $\xi = \text{radiation efficiency, } 0 < \xi \leq 1$                                                                            |
| $G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$                                                                                        |
| <i>For lossless / ideal antennas: <math>G = D</math>.</i>                                                                               |

COMMON DIPOLE PARAMETERS

| Type          | $D$  | $D_{\text{dB}}$ | $R_{\text{rad}} \text{ (}\Omega\text{)}$ |
|---------------|------|-----------------|------------------------------------------|
| Isotropic     | 1    | 0               | —                                        |
| Hertzian      | 1.5  | 1.76            | $80\pi^2 (l/\lambda)^2$                  |
| Half-wave     | 1.64 | 2.15            | 73                                       |
| $l = \lambda$ | 2.41 | 3.82            | 199                                      |

EFFECTIVE AREA  $A_e$

|                                                                                   |
|-----------------------------------------------------------------------------------|
| Definition $\text{m}^2$ , <i>antenna's receiving cross section</i>                |
| $A_e = \frac{\lambda^2 D}{4\pi}$                                                  |
| Physical cross section <i>half-wave dipole, wire diameter <math>d</math></i>      |
| $A_p = l d = \frac{\lambda}{2} d$                                                 |
| Ratio $A_e \gg A_p$ <i>for thin wires (good!)</i>                                 |
| Example: $f = 100 \text{ MHz}, d = 2 \text{ cm} \Rightarrow A_e/A_p \approx 39$ . |

POWER FLOW IDENTITIES

|                                                                                    |
|------------------------------------------------------------------------------------|
| $P_{\text{rad}} = S_{\text{max}} \Omega_p R^2 = \frac{4\pi R^2 S_{\text{max}}}{D}$ |
| $S_{\text{max}} = \frac{P_{\text{rad}} D}{4\pi R^2} \text{ (W/m}^2\text{)}$        |
| <i>For isotropic radiator: <math>S = P_{\text{rad}}/(4\pi R^2)</math>.</i>         |

FRIIS TRANSMISSION FORMULA

|                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|
| Far-field link budget $P_t \text{ tx, } P_r \text{ rx, dist } R$                                                               |
| $P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t$                                                                    |
| $G_t, G_r$ linear gains; $\lambda = c/f$ ; $R$ in m; $P_t, P_r$ in W.                                                          |
| Solve for $R$                                                                                                                  |
| $R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$                                                                      |
| Solve for $G_r$ given $A_e$                                                                                                    |
| $G_r = \frac{4\pi A_{e,r}}{\lambda^2} \Rightarrow P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$                                      |
| <i>Use linear <math>G</math>, not dB. Convert first: <math>G_{\text{lin}} = 10^{G_{\text{dB}}/10}</math>.</i>                  |
| <i>Free-space path loss <math>L_{\text{fs}} = (4\pi R/\lambda)^2</math> so <math>P_r = G_t G_r P_t / L_{\text{fs}}</math>.</i> |

dB  $\leftrightarrow$  LINEAR

|                                                                                            |
|--------------------------------------------------------------------------------------------|
| Power gain <i>factor <math>\rightarrow</math> dB</i>                                       |
| $G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$ |
| dB linear    dB linear                                                                     |
| 0    1    10    10                                                                         |
| 3    2    13    20                                                                         |
| 6    4    20    100                                                                        |
| 7    5    30    1000                                                                       |
| −3    0.5    −10    0.1                                                                    |
| <i>For <math>E</math> or amplitude use <math>20 \log</math> not <math>10 \log</math>.</i>  |

RADAR EQUATION (BONUS)

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| Monostatic radar $\text{tx} = \text{rx, target cross section } \sigma_t \text{ (m}^2\text{)}$   |
| $P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}$                                         |
| Doppler shift <i>closing velocity <math>v_r</math></i>                                          |
| $f_d = \frac{2v_r}{\lambda} = \frac{2v_r f}{c} \text{ (Hz)}$                                    |
| <i>Factor of 2 from round-trip path; <math>v_r</math> is the radial component toward radar.</i> |

LECTURE-11 BOUNDARY  $\leftrightarrow$  T-LINE

| EM            | T-line          | Match          |
|---------------|-----------------|----------------|
| $\tilde{E}_i$ | $\tilde{V}_1^+$ | forward wave   |
| $\tilde{E}_r$ | $\tilde{V}_1^-$ | reverse wave   |
| $\eta_i$      | $Z_{0i}$        | line impedance |
| $\Gamma$      | $\Gamma$        | same formula   |
| $k_i$         | $\beta_i$       | wavenumber     |

*Bouncing-wave analogy: every t-line formula in Ch. 2 transfers verbatim with the substitutions above.*

ANTENNA INTEGRATION TIPS

|                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|
| <i>Many radiation integrals (e.g. <math>\Omega_p</math> for <math>e^{-20\theta^2} \sin \theta</math>) have no closed form.</i> |
| <i>Use Wolfram / numerical integration. State this in the answer.</i>                                                          |
| <i>Common shortcut: if <math>F(\theta)</math> is narrow (<math>\theta_{1/2} \ll 1</math>), approximate</i>                     |
| $\Omega_p \approx \theta_{1/2}^2 \pi / (4 \ln 2) \text{ (Gaussian beam)}$                                                      |

DIPOLE

|                                                |
|------------------------------------------------|
| $l$ — antenna length (m)                       |
| $\lambda = c/f$ — wavelength (m)               |
| $I_0$ — peak current (A)                       |
| $\eta_0 = 120\pi \approx 377 \Omega$           |
| $k = 2\pi/\lambda$ — wavenumber (rad/m)        |
| $R_{\text{rad}}$ — radiation res. ( $\Omega$ ) |
| $R$ — obs. dist (m)                            |

RADIATION

|                                                   |
|---------------------------------------------------|
| $S(R, \theta)$ — power density ( $\text{W/m}^2$ ) |
| $S_{\text{max}}$ — peak power density             |
| $F(\theta) = S/S_{\text{max}}$ — normalized       |
| $\Omega_p$ — pattern solid angle (sr)             |
| $\beta$ — HPBW (rad or deg)                       |
| $D$ — directivity (unitless)                      |
| $P_{\text{rad}}$ — total rad. power (W)           |

ANTENNA

|                                                   |
|---------------------------------------------------|
| $A_e = \lambda^2 D / (4\pi)$ — effective area     |
| $A_p$ — physical cross sec. ( $\text{m}^2$ )      |
| $\xi$ — rad. efficiency ( $0 < \xi \leq 1$ )      |
| $G = \xi D$ — gain (unitless)                     |
| $G_{\text{dB}} = 10 \log G$                       |
| Half-wave: $D = 1.64, R_{\text{rad}} = 73 \Omega$ |
| Hertzian: $D = 1.5$                               |

FRIIS / LINK

|                                                  |
|--------------------------------------------------|
| $P_t$ — tx power (W)                             |
| $P_r$ — rx power (W)                             |
| $G_t, G_r$ — linear gains                        |
| $R$ — range (m)                                  |
| $L_{\text{fs}} = (4\pi R/\lambda)^2$ — path loss |
| 3 dB $\rightarrow$ 2; 10 dB $\rightarrow$ 10     |
| $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$         |

RADAR

|                                   |
|-----------------------------------|
| $\sigma_t$ — RCS ( $\text{m}^2$ ) |
| $f_d$ — Doppler shift (Hz)        |
| $v_r$ — radial vel. (m/s)         |
| $P_r \propto 1/R^4$ (monostatic)  |
| $f_d = 2v_r/\lambda$ (round trip) |
| $c = 3 \times 10^8 \text{ m/s}$   |
| $d > 0$ : target approaching      |